

Instruction: Delete any examples in grey italic text that are not relevant/applicable to the study once the programming specification has been finalized

Section A: Data Source Selection		
Data source	Population Subset	Version
Optum	Clinformatics Data Mart (CDM)	2025q2

Section B: Define Study Time Periods			
Study Time Period	Label	Definition	Additional Notes
STUDY PD	Full study period	01Jul2015 - 30June2025 (inclusive)	Full study time period
ID PD	Patient identification period	01Jan2016 - 30June2025 (inclusive)	2

INDEX	Index date	Date of the patient's earliest qualifying multiple myeloma diagnosis date	The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in tab 42 "MM dx code".	OK19012026
			Flags should also be made to distinguish which claims are non-diagnostic claims. A non-diagnostic claim refers to a claim for treatment of a condition (e.g., infusion) rather than diagnosis (see tab 49 "Diagnostic").	
			Patients must have six months of CE with medical or pharmacy benefits before the index date. Patients with gaps in enrollment of < 30 days are considered to be continuously enrolled.	
Baseline	Baseline period	The baseline period is fixed and is the six months before the index date.	Patients are only required to have ≥ 1 day of CE starting on the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled; consider medical and/or pharmacy benefits. Follow up period ends the earliest of death or the end of the study period (30June2025).	OK19012026
FU PD	Follow up period	The follow up period is fixed and is from the index date till the end of the patient identification period (inclusive)	Note: Patients with ≥ 3 months of CE with medical or pharmacy benefits from the index date (with no allowable gaps in enrollment) will be flagged for consideration for sensitivity or secondary analyses, and ending on the earliest of health plan disenrollment or the study period for patients who do not have have an assigned date of death < 3mos after the index date.	OK19012026
e.g., PS124	e.g., 24-month post-index period	e.g., INDEX_DT through INDEX_DT + 730		
etc				

Instructions: Section C should ONLY be used to document decisions to the source data before the analysis file build. If decisions apply to a variable in the analysis file, document in the appropriate variable definition.

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25	#	Source Table/Variable	Situation Requiring Resolution	Action	Reasoning / Comments	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)								
26	1	Days to months conversion	Assuming 30.4375 day per month	3 months = 91 days, 6 months=183 days, 9 months=274 days, 12 months=365 days	Consistent with derivation other GSK Oncology studies		OK 19012026								
27	2	ELIG_COH	Study team does not want to exclude smoldering MM patients. Optum built study cohort with the intention to exclude smoldering MM patients.	The team will additionally include patients who had ≥1 non-diagnostic medical claim for MM during the baseline period. These patients were excluded by Optum originally. If a patient has newly diagnosed or active MM, it would be likely that they would experience frequent activity, hence why we look for 2 non-diagnostic MM claims close together in the first step of the attrition table. A smoldering patient would have infrequent activity, hence why a claim for MM that falls outside of the 30, 60 or 90 days window requirement that's used to identify newly diagnosed patients is likely a smoldering patient.	Study team will consider excluding patients with a stand alone treatment code (rather than any non-diagnostic code) on an MM claim in the baseline period for future sensitivity analysis. This would not remove smoldering patients but rather ensure our patients were <u>newly</u> active MM patients.		OK 19012026								
28	3	FU_PD	Original IE criteria from Optum only considers patients with at least 6mos of CE from index onwards.	Study team wants to keep criteria as broad as possible.	Will only require at least a day of CE post index, though will flag patients with 3 or more months of CE, to potentially subset on these patients for later consideration.		OK 19012026								
29	4	MED_COND	Conditioning medications are included in MM-approved code list Approach 1: Use TOS_CD if conditions are not restricted on inpatient services associated with a hospitalization which requires an admission and discharge date. 1. Extract the records from T_Medical and T_CONFINEMENT datasets Inpatient -> POS IN (21, 51, 61) or TOS_CD IN (FAC_IP ACUTE, FAC_IP REHSNF, PROF_INPVIS, FAC_IP SNF) Outpatient->none of the above Approach 2: For anything related to services happening during a hospitalization with room and board, apply the condition on the CONF_ID because the majority of the work we do includes both diagnoses and medical services e.g. procedures, and when we talk about inpatient utilization, it is usually in the context of an admission and discharge. 1. Inpatient records should be restricted to cases where CONF_ID is not NULL, from the T_CONFINEMENT table, where this has associated admission and discharge dates. 2. The confinement table has summary information, so if users are pulling medical claims for specific inclusion, user should match on the CONF_ID variable, where it is not NULL in the confinement table. All other records without a CONF_ID or where CONF_ID is NULL should be considered non-inpatient.	Ignore these treatments for now.	The team is not particularly interested in conditioning therapy. These treatments potentially will be added back to the LOT at a later time.		OK 19012026								
30	5	Identifying inpatient and non-inpatient claims	Approach: We implement both Approach 1 and Approach 2 in combination (OR logic) per Optum CDM business rules guidance. Approach 1 (POS/TOS): Inpatient if POS IN (21, 51, 61) OR TOS_CD IN (FAC_IP ACUTE, FAC_IP REHSNF, PROF_INPVIS, FAC_IP SNF) Approach 2 (CONF_ID): Inpatient if CONF_ID is NOT NULL and validated against T_CONFINEMENT table with both ADMIT_DATE and DISCH_DATE present Non-inpatient: All records not meeting either approach above A claim is classified as inpatient if either approach identifies it as inpatient. This maximizes capture of true inpatient encounters.	For this LOT algorithm, the team has decided not to define	Rationale: Using both approaches together provides the most comprehensive inpatient identification. From our validation on the 2025q2 data: Total patients (Step 0): 94951 Any inpatient event: 43439 (45.2%) Via POS/TOS (Appr 1): 42618 (44.3%) Via CONF_ID (Appr 2): 40357 (42.0%) Both approaches: 39536 (41.1%) POS/TOS only: 3082 (3.2%) CONF_ID only: 821 (0.9%) Outpatient only: 52757 (54.8%)		OK 19012026								
<	>	1.Cover	2.QC Review	3.Data Prep	4.Study/Pop	5A.MMA.MED	5B.MAP.MED	6.LOTT.BASE	7.SCT	10.LOTT.BASE.END.OK	11.LOTT.2.BASE	12.LOTT.5.BASE.END	13.SCT.CART.LOTT.START	14.LOTT.2.Scenarios	40.Cl.MMA.RC

A	B	C	D	E	F	G
6	Defining maintenance periods	Defining mtx periods is quite complex, especially with the elimination of the 90 day gap rule originally proposed by Optum. Without allowable (90d) gaps, for which mtx periods relied on for defining starts and stops, and due to the messy nature of claims and prescriber behavior, it is very difficult to create an overarching mtx rule.	For this LOT algorithm, the team has decided not to define maintenance therapy as its own LOT but rather to flag the inclusion of LOT1 regimens that include potential valid maintenance therapies/therapy combinations.	Reporting mtx is not of importance/relevance to upcoming studies planned in Multiple Myeloma. Flags will enable users to define mtx themselves in the future if so needed.		
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